Model routing: one-page cheat sheet

The decision

Filter by capabilities and policy, then choose a model before generation. A route is useful only under a stated quality requirement, cost objective and latency budget. A cascade may make multiple calls and must count every attempt.

Implemented offline policies

Rules: complexity keyword, then length >180 characters, otherwise small. Local: word-count naive Bayes trained on 32 authored prompts. Default large-class threshold: 0.6. An ambiguity band of +/-0.05 or zero known words selects large. The score is not answer confidence. Both receive prompt text only.

Run from the code folder

```
python3 -m unittest discover -s tests -v
python3 -m routing_workshop verify
python3 -m routing_workshop route "Prove it." --policy local
python3 -m routing_workshop benchmark --mode synthetic
python3 -m routing_workshop benchmark --mode recorded
python3 -m scripts.experiment
```

Evidence labels

SYNTHETIC: designed hidden success outcomes, no LLM inference. 32 train / 12 dev / 36 test prompts. Freeze choices on dev before test. RECORDED REPLAY: 24 historical MT-Bench response pairs with exact-source checks. Quality is human pairwise preference credit, ties = half. It is not accuracy or current model performance. In BOTH modes, service time and cost units are assumptions. Only local router/dispatch overhead is measured. No live API adapter ships.

A fair report

Use identical prompts and per-model outcomes for fixed, random and learned policies. Count each unique question once for quality. Include routing, dispatch and service in end-to-end time. Report p50/p95, quality, assumed or actual cost, failure examples and uncertainty limits. Replay lookup time is not inference latency.

One experiment to take home

Write an objective. Add short difficult, long simple and quoted-keyword cases. Split before tuning. Freeze the policy, evaluate once, inspect errors and state what the evidence cannot establish.

Python 3.10+ only. No keys, GPU, model download or paid API required. See README.md for commands and handouts/ for the worksheet and take-home tasks.